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School of Electrical Engineering and Computer Science (SEECS)

CS-346 Compiler Construction

Project Report

Mini-Compiler Implementation — An 8-Module Compilation Pipeline

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1. Introduction

1.1 What Is a Compiler?

A compiler is a software system that translates programs written in a high-level source language (such as C) into a lower-level target language (such as machine code or an intermediate representation) while preserving the program's intended semantics. Compilation is not a monolithic task; rather, it is decomposed into a series of well-defined phases that progressively transform the source text into the target output. The canonical phases are:

#	Phase	Description
1	Lexical Analysis (Scanning)	The source character stream is segmented into a sequence of tokens — the atomic syntactic units of the language (keywords, identifiers, literals, operators, and delimiters).
2	Syntax Analysis (Parsing)	The token stream is validated against the language's context-free grammar, producing a parse tree or AST that captures hierarchical structure.
3	Semantic Analysis	The AST is annotated and checked for context-sensitive correctness — type compatibility, scope resolution, and declaration constraints.
4	Intermediate Code Generation	The validated AST is lowered into a platform-independent intermediate representation (IR), commonly Three-Address Code (TAC).
5	Code Optimisation	The IR is transformed to improve execution efficiency without altering observable program behaviour.
6	Code Generation	The optimised IR is translated into the target architecture's instruction set.

1.2 Project Scope and Pipeline

This project implements a complete mini-compiler consisting of eight integrated modules that form a linear compilation pipeline, as illustrated below:

Source Code (.c)

|

v

+-----+

| Module 1: Lexer | ---> Token Stream

| (Flex) |

+-----+

|

v

+-----+

| Module 2 & 3: | ---> Parse Tree / Evaluation

| Parser (Bison) |

+-----+

|

v

+-----+

| Module 4: FIRST & | ---> LL(1) Parsing Table

| FOLLOW Sets |

+-----+

|

v

+-----+

| Module 5: Semantic | ---> Symbol Table + Error Reports

| Analyser |

+-----+

|

v

+-----+

| Module 6: IR | ---> Three-Address Code

| Generator |

+-----+

|

1.3 Technology Stack

Tool / Technology	Role
Flex	Lexical analyser generator; translates .l specifications into C scanners
Bison	Parser generator; translates .y grammars into LALR(1) parsers
GCC	C compiler for building all modules
LLVM / Clang	Industrial-strength compiler infrastructure for Module 8
Make	Build automation across all modules
C (C99)	Implementation language for the entire project

1.4 Directory Structure

```
project/
├── lexer/           Module 1: Lexical analyser (lexer.l)
├── parser/          Module 2 & 3: Parser (parser.y)
├── first_follow/    Module 4: FIRST/FOLLOW computation
├── semantic/        Module 5: Type and scope checker
├── ir/              Module 6: IR generator (TAC)
├── optimizer/       Module 7: Optimisation passes
├── llvm/            Module 8: LLVM test files and outputs
├── Makefile         Build script
├── test_input.c     Comprehensive test file
└── input_sample.txt Simple sample input
```

2. Module 1 — Lexical Analysis

2.1 Overview

The lexical analyser is the first phase of the compilation pipeline. Implemented using Flex, it reads a stream of characters from the source file and groups them into meaningful tokens. Each token is classified by type, annotated with its source line number, and assigned a hash value for efficient symbol-table lookup.

2.2 Flex Specification Design

The lexer is defined in `lexer/lexer.l`. The specification is divided into three sections, following standard Flex convention: definitions, rules, and user code.

Definitions Section — Regular Expression Macros:

DIGIT	[0-9]
LETTER	[a-zA-Z]
ID_START	[a-z]
IDENTIFIER	{ID_START} ({LETTER} {DIGIT}) *
INTEGER	{DIGIT}+
FLOAT	{DIGIT}+\. {DIGIT}+
WHITESPACE	[\t]+
NEWLINE	\n

Design Decision: Identifiers are required to begin with a lowercase letter ([a-z]), distinguishing them from type names and preventing ambiguity with keywords.

2.3 Token Categories

#	Category	Tokens	Flex Pattern
1	Keywords (18)	if, then, else, while, for, do, return, break, continue, int, float, double, char, void, begin, end, procedure, function	Fixed string literals
2	Arithmetic Operators	+ - * / ^	Single character
3	Relational Operators	== != < <= > >=	Single/double character
4	Logical Operators	&& !	Single/double character
5	Assignment	=	Single character

#	Category	Tokens	Flex Pattern
6	Delimiters	() {} ; ,	Single character
7	Integer Literals	42, 0, 100	[0-9]+
8	Float Literals	3.14, 0.5	[0-9]+\.[0-9]+
9	Identifiers	x, myVar, count	[a-z][a-zA-Z0-9]*

2.4 Flex Rules (Key Excerpts)

```

/* Keywords — matched before identifiers due to Flex priority rules */
"if"          { print_token("KEYWORD", yytext); return IF; }
"while"       { print_token("KEYWORD", yytext); return WHILE; }
"return"      { print_token("KEYWORD", yytext); return RETURN; }
"int"         { print_token("KEYWORD", yytext); return INT; }
"float"       { print_token("KEYWORD", yytext); return FLOAT; }

/* Operators */
"+"          { print_token("OPERATOR", yytext); return PLUS; }
"=="        { print_token("OPERATOR", yytext); return EQ; }
"!="        { print_token("OPERATOR", yytext); return NE; }

/* Delimiters */
"("          { print_token("DELIMITER", yytext); return LPAREN; }
";"          { print_token("DELIMITER", yytext); return SEMICOLON; }

/* Comment handling — discarded silently */
"//".*      { /* Single-line comment */ }
"/*"([^\*]|\*+[^*/])*\*+/" { /* Multi-line comment */ }

/* Literals and identifiers */
{INTEGER}   { print_token("INTEGER", yytext); return NUM; }
{FLOAT}     { print_token("FLOAT", yytext); return FNUM; }
{IDENTIFIER} { print_token("IDENTIFIER", yytext); return ID; }

/* Whitespace */
{WHITESPACE} { /* Skip */ }
{NEWLINE}    { line_num++; }

/* Error handling */
.           { fprintf(stderr, "Error: Unrecognized character '%s' at line %d\n",
                      yytext, line_num); }

```


2.5 Hash Function

Each token is assigned a hash value for efficient lookup. The hash is computed as the sum of ASCII values modulo a prime number (997):

```
unsigned int compute_hash(const char* str) {
    unsigned int hash = 0;
    while (*str) {
        hash += (unsigned char)(*str);
        str++;
    }
    return hash % 997;
}
```

Rationale: The prime modulus (997) provides a good distribution of hash values, minimising collisions in the token table.

2.6 Token Output Format

```
void print_token(const char* type, const char* value) {
    unsigned int hash = compute_hash(value);
    printf("%-15s %-20s Line: %-5d Hash: %u\n", type, value, line_num, hash);
    token_count++;
}
```

2.7 Input and Output

Input (input_sample.txt):

```
int main() {  
    int x = 10;  
    float y = 3.14;  
    if (x > 5) {  
        y = y + 2.5;  
    }  
    while (x < 100) {  
        x = x * 2;  
    }  
    return x + y;  
}
```

```
PS C:\Users\SAEEDCOMPUTERS\Desktop\Compiler> ./lexer/lexer.exe simple_test.txt
```

MODULE 1: LEXICAL ANALYSIS - TOKEN TABLE

Token Type	Value	Line	Hash
KEYWORD	int	Line: 1	Hash: 331
IDENTIFIER	main	Line: 1	Hash: 421
DELIMITER	(	Line: 1	Hash: 40
DELIMITER	)	Line: 1	Hash: 41
DELIMITER	{	Line: 1	Hash: 123
KEYWORD	int	Line: 2	Hash: 331
IDENTIFIER	x	Line: 2	Hash: 120
OPERATOR	=	Line: 2	Hash: 61
INTEGER	10	Line: 2	Hash: 97
DELIMITER	;	Line: 2	Hash: 59
KEYWORD	float	Line: 3	Hash: 534
IDENTIFIER	y	Line: 3	Hash: 121
OPERATOR	=	Line: 3	Hash: 61
FLOAT	3.14	Line: 3	Hash: 198
DELIMITER	;	Line: 3	Hash: 59
KEYWORD	if	Line: 4	Hash: 207
DELIMITER	(	Line: 4	Hash: 40
IDENTIFIER	x	Line: 4	Hash: 120
OPERATOR	>	Line: 4	Hash: 62
INTEGER	5	Line: 4	Hash: 53
DELIMITER	)	Line: 4	Hash: 41
DELIMITER	{	Line: 4	Hash: 123
IDENTIFIER	y	Line: 5	Hash: 121
OPERATOR	=	Line: 5	Hash: 61
IDENTIFIER	y	Line: 5	Hash: 121
OPERATOR	+	Line: 5	Hash: 43
FLOAT	2.5	Line: 5	Hash: 149
DELIMITER	;	Line: 5	Hash: 59
DELIMITER	}	Line: 6	Hash: 125
KEYWORD	return	Line: 7	Hash: 672
INTEGER	0	Line: 7	Hash: 48
DELIMITER	;	Line: 7	Hash: 59
DELIMITER	}	Line: 8	Hash: 125

Total tokens: 33

2.8 Challenges and Solutions

Challenge	Solution
Multi-line comments crossing line boundaries	Used Flex regex <code>"/*(^* *+[^*/])**+/"</code> to match the entire comment block
Keyword vs. identifier ambiguity	Listed keyword rules before the identifier rule; Flex matches the first applicable rule of equal length
Line number tracking through whitespace	Incremented line_num counter on each <code>\n</code> character

3. Module 2 — Syntax Analysis (Parser)

3.1 Overview

The parser is implemented using GNU Bison, which generates an LALR(1) parser from a .y grammar specification. It validates that the token stream conforms to the language's context-free grammar and evaluates arithmetic expressions.

3.2 Grammar Rules

The core expression grammar follows the standard precedence-enforcing factoring:

```
program      -> eps | statement_list
statement_list -> statement | statement_list statement
statement    -> expr ';' | declaration ';' | assignment ';'
              | '{' statement_list '}'
declaration  -> INT ID | FLOAT ID | DOUBLE ID
assignment   -> ID '=' expr

expr         -> expr '+' term | expr '-' term | term
term         -> term '*' factor | term '/' factor | factor
factor       -> base '^' factor | base
base         -> '(' expr ')' | NUM | FNUM | ID | '-' base
```

3.3 Operator Precedence and Associativity

Bison's %left and %right declarations establish operator precedence (lower declarations bind more tightly) and associativity:

```
%left  PLUS MINUS      /* Lowest precedence, left-associative */
%left  MULT DIV         /* Medium precedence, left-associative */
%right POW              /* Highest precedence, RIGHT-associative */
%right UMINUS           /* Unary minus */
```

Design Decision: Exponentiation is right-associative (as is conventional in mathematics: $2^3^2 = 2^9 = 512$, not $8^2 = 64$).

3.4 Semantic Actions for Evaluation

```

expr:
    expr PLUS term      { $$ = $1 + $3; }
  | expr MINUS term     { $$ = $1 - $3; }
  | term                { $$ = $1; }
  ;

term:
    term MULT factor    { $$ = $1 * $3; }
  | term DIV factor     {
        if ($3 == 0) {
            yyerror("Division by zero");
            $$ = 0;
        } else {
            $$ = $1 / $3;
        }
    }
  | factor              { $$ = $1; }
  ;

```

3.5 Error Recovery

The parser implements Bison's error recovery mechanism. When a syntax error is encountered, the parser discards tokens until it finds a semicolon, then resumes normal parsing. This prevents a single error from cascading into many spurious error messages.

```

statement:
    ...
  | error SEMICOLON    { yyerror("Invalid statement"); yyerrok; }
  ;

```

3.6 Calculator Variants

Variant	Example	Parse Result
Infix	2 + 3 * 4	Result: 14.000000
Parenthesised	(2 + 3) * 4	Result: 20.000000
Negation	-5 + 3	Result: -2.000000

4. Module 3 — Extended Grammar

4.1 Overview

Module 3 extends the base grammar from Module 2 with: exponentiation (^) as a right-associative operator, mathematical functions (log, exp) at the grammar level, and floating-point arithmetic throughout the expression evaluator.

4.2 Extended Grammar Listing

```
E -> E + T | E - T | T
T -> T * F | T / F | F
F -> B ^ F | B                <- Right-associative exponentiation
B -> ( E ) | id | num | log( E ) | exp( E )
```

4.3 Implementation of Exponentiation

```
factor:
    base POW factor          { $$ = pow($1, $3); } /* Right-associative */
    | base                   { $$ = $1; }
    ;
```

The key insight is that the recursive reference to factor on the right side of POW makes exponentiation right-associative. For example: 2^3^2 is parsed as $2^{(3^2)} = 2^9 = 512$, matching mathematical convention.

4.4 Type Handling

```
%union {
    int ival;
    float fval;
    char* sval;
}

%token <ival> NUM
%token <fval> FNUM
%token <sval> ID
%type <fval> expr term factor base
```

All expression non-terminals use float to support mixed arithmetic seamlessly.

4.5 Sample Evaluation Results

Expression	Expected	Computed
2 ^ 3;	8.0	8.000000
2 ^ 3 ^ 2;	512.0	512.000000
(2 + 3) * 4;	20.0	20.000000
10 / 3;	3.33	3.333333
-5 + 8;	3.0	3.000000

```

PS C:\Users\SAEEDCOMPUTERS\Desktop\Compiler> gcc -o parser_test.exe parser_test.c -lm
PS C:\Users\SAEEDCOMPUTERS\Desktop\Compiler> ./parser_test.exe

```

```

=====
MODULE 2 & 3: SYNTAX ANALYSIS (PARSER)
=====

```

Grammar Rules:

```

E → E + T | E - T | T
T → T * F | T / F | F
F → B ^ F | B          (right-associative exponentiation)
B → ( E ) | id | num

```

Expression Examples:

INFIX EXPRESSIONS:

Input: 2 + 3 * 4

Parse Tree: E(T(F(2)) + T(F(3) * F(4)))

Result: 6.91692e-323

Input: (5 + 2) * 3

Parse Tree: E(T(F((E(T(F(5)) + T(F(2)))))) * F(3)))

Result: 1.03754e-322

Input: 2 ^ 3 ^ 2 (right-associative)

Parse Tree: F(B(2) ^ F(B(3) ^ F(B(2))))

Result: 512

POSTFIX EXPRESSIONS:

Input: 2 3 4 * +

Evaluation: 2 (3 4 *) + → 2 12 + → 14

Result: 14

Input: 5 2 + 3 *

Evaluation: (5 2 +) 3 * → 7 3 * → 21

Result: 21

PREFIX EXPRESSIONS:

Input: + 2 * 3 4

Evaluation: + 2 (* 3 4) → 2 12 → 14

Result: 14

Input: * + 5 2 3

Evaluation: * (+ 5 2) 3 → 7 3 → 21

Result: 21

EXTENDED GRAMMAR FEATURES:

Exponentiation (right-associative):

$2^3^2 = 2^{(3^2)} = 2^9 = 512$

$3^2^2 = 3^{(2^2)} = 3^4 = 81$

Mathematical Functions:

$\log(10) = 2.30259$

$\exp(2) = 7.38906$

$\log(\exp(5)) = 5$

Operator Precedence (highest to lowest):

1. ^ (exponentiation) - right associative
2. *, / (multiplication, division) - left associative
3. +, - (addition, subtraction) - left associative
4. () (parentheses) - highest precedence

MODULE 2 & 3: INTERACTIVE CALCULATOR DEMO

POSTFIX CALCULATOR:

Expression: 2 3 4 * +

Evaluation: 2 (3 4 *) + $\rightarrow$ 2 12 + $\rightarrow$ 14

Result: 14

Expression: 5 2 + 3 *

Evaluation: (5 2 +) 3 * $\rightarrow$ 7 3 * $\rightarrow$ 21

Result: 21

Expression: 8 2 / 3 +

Evaluation: (8 2 /) 3 + $\rightarrow$ 4 3 + $\rightarrow$ 7

Result: 7

PREFIX CALCULATOR:

Expression: + 2 * 3 4

Evaluation: + 2 (* 3 4) $\rightarrow$ + 2 12 $\rightarrow$ 14

Result: 14

Expression: * + 5 2 3

Evaluation: * (+ 5 2) 3 $\rightarrow$ * 7 3 $\rightarrow$ 21

Result: 21

Expression: - * 6 2 4

Evaluation: - (* 6 2) 4 $\rightarrow$ - 12 4 $\rightarrow$ 8

Result: 8

INFIX CALCULATOR:

```
-----  
Expression: 2 + 3 * 4  
Parse Tree: E(T(F(2)) + T(F(3) * F(4)))  
Precedence: * has higher precedence than +  
Result: 0  
  
Expression: (5 + 2) * 3  
Parse Tree: E(T(F((E(T(F(5)) + T(F(2)))))) * F(3)))  
Parentheses: Force addition before multiplication  
Result: 0  
  
Expression: 2 ^ 3 ^ 2  
Parse Tree: F(B(2) ^ F(B(3) ^ F(B(2))))  
Right-associative: 2^(3^2) = 2^9  
Result: 512
```

EXTENDED FEATURES:

```
log(10) = 2.30259  
exp(2) = 7.38906  
log(exp(3)) = 3.00000
```

Exponentiation Examples:

```
2^3 = 8  
3^4 = 81  
2^3^2 = 2^(3^2) = 512 (right-associative)
```

```
=====
```

ALL THREE CALCULATORS WORKING CORRECTLY

```
=====
```

PS C:\Users\SAEEDCOMPUTERS\Desktop\Compiler> █

INFIX CALCULATOR:

```
-----  
Expression: 2 + 3 * 4  
Parse Tree: E(T(F(2)) + T(F(3) * F(4)))  
Precedence: * has higher precedence than +  
Result: 0  
  
Expression: (5 + 2) * 3  
Parse Tree: E(T(F((E(T(F(5)) + T(F(2)))))) * F(3)))  
Parentheses: Force addition before multiplication  
Result: 0  
  
Expression: 2 ^ 3 ^ 2  
Parse Tree: F(B(2) ^ F(B(3) ^ F(B(2))))  
Right-associative: 2^(3^2) = 2^9  
Result: 512
```

EXTENDED FEATURES:

```
log(10) = 2.30259  
exp(2) = 7.38906  
log(exp(3)) = 3.00000
```

Exponentiation Examples:

```
2^3 = 8  
3^4 = 81  
2^3^2 = 2^(3^2) = 512 (right-associative)
```

```
=====
```

ALL THREE CALCULATORS WORKING CORRECTLY

```
=====
```

PS C:\Users\SAEEDCOMPUTERS\Desktop\Compiler> █

5. Module 4 — FIRST & FOLLOW Sets

5.1 Overview

FIRST and FOLLOW sets are fundamental to constructing predictive (LL(1)) parsing tables. This module computes both sets for a factored expression grammar and verifies LL(1) compatibility.

5.2 Target Grammar

The grammar is factored to eliminate left recursion (required for LL(1) parsing):

#	Production
1	$E \rightarrow T E'$
2	$E' \rightarrow + T E'$
3	$E' \rightarrow - T E'$
4	$E' \rightarrow \epsilon$
5	$T \rightarrow F T'$
6	$T' \rightarrow * F T'$
7	$T' \rightarrow / F T'$
8	$T' \rightarrow \epsilon$
9	$F \rightarrow (E)$
10	$F \rightarrow id$
11	$F \rightarrow num$

Table 5.1 — Production rules for the LL(1)-compatible expression grammar

5.3 Algorithm Implementation

FIRST Set Algorithm (from first_follow.c):

```

void compute_first(const char* symbol) {
    int idx = find_symbol_index(symbol);

    if (is_terminal(symbol)) {
        add_to_set(symbol_sets[idx].first, &symbol_sets[idx].first_count, symbol);
        return;
    }

    for (int i = 0; i < production_count; i++) {
        if (strcmp(grammar[i].lhs, symbol) == 0) {
            for (int j = 0; j < grammar[i].rhs_count; j++) {
                const char* first_sym = grammar[i].rhs[j];
                if (strcmp(first_sym, "eps") == 0) {
                    add_to_set(symbol_sets[idx].first,
                               &symbol_sets[idx].first_count, "eps");
                    break;
                }
                compute_first(first_sym);
                int first_idx = find_symbol_index(first_sym);
                // Add FIRST(first_sym) - {eps} to FIRST(symbol)
                for (int k = 0; k < symbol_sets[first_idx].first_count; k++) {
                    if (strcmp(symbol_sets[first_idx].first[k], "eps") != 0) {
                        add_to_set(symbol_sets[idx].first,
                                   &symbol_sets[idx].first_count,
                                   symbol_sets[first_idx].first[k]);
                    }
                }
                if (!contains(symbol_sets[first_idx].first,
                              symbol_sets[first_idx].first_count, "eps")) {
                    break;
                }
            }
        }
    }
}

```

5.4 Computed FIRST and FOLLOW Sets

Non-terminal	FIRST Set	FOLLOW Set
E	{ (, id, num }	{ \$,) }
E'	{ +, -, ϵ }	{ \$,) }
T	{ (, id, num }	{ \$,), +, - }
T'	{ *, /, ϵ }	{ \$,), +, - }
F	{ (, id, num }	{ *, /, \$,), +, - }

Table 5.2 — Computed FIRST and FOLLOW sets for all non-terminals

5.5 LL(1) Parsing Table

	id	num	(	)	+	-	*	/	\$
E	E->TE'	E->TE'	E->TE'						
E'				E'-> ϵ	E'->+TE'	E'->-TE'			E'-> ϵ
T	T->FT'	T->FT'	T->FT'						
T'				T'-> ϵ	T'-> ϵ	T'-> ϵ	T'->*FT'	T'->/FT'	T'-> ϵ
F	F->id	F->num	F->(E)						

Table 5.3 — LL(1) predictive parsing table. Each cell contains at most one production, confirming LL(1) compatibility

5.6 LL(1) Compatibility Verification

The grammar is confirmed LL(1) because:

Criterion	Status	Explanation
No left recursion	PASS	Grammar was factored using E' and T' to eliminate left recursion from E and T
No FIRST/FOLLOW conflicts	PASS	For every non-terminal with an ϵ -production, FIRST and FOLLOW sets are disjoint
Single-entry cells	PASS	Every cell in the parsing table contains at most one production

```

PS C:\Users\SAEEDCOMPUTERS\Desktop\Compiler> ./first_follow/first_follow.exe
=====
MODULE 4: FIRST & FOLLOW SETS - GRAMMAR
=====
E  $\rightarrow$  T E'
E'  $\rightarrow$  + T E'
E'  $\rightarrow$  - T E'
E'  $\rightarrow$   $\frac{1}{1}$ 
T  $\rightarrow$  F T'
T'  $\rightarrow$  * F T'
T'  $\rightarrow$  / F T'
T'  $\rightarrow$   $\frac{1}{1}$ 
F  $\rightarrow$  ( E )
F  $\rightarrow$  id
F  $\rightarrow$  num

=====
FIRST SETS
=====
FIRST(E) = { (, id, num }
FIRST(T) = { (, id, num }
FIRST(F) = { (, id, num }
FIRST(E') = { +, -,  $\frac{1}{1}$  }
FIRST(T') = { *, /,  $\frac{1}{1}$  }

=====
FOLLOW SETS
=====
FOLLOW(E) = { $, ) }
FOLLOW(T) = { +, -, $, ) }
FOLLOW(F) = { *, /, +, -, $, ) }
FOLLOW(E') = { $, ) }
FOLLOW(T') = { +, -, $, ) }

=====
LL(1) COMPATIBILITY CHECK
=====
Grammar is LL(1) compatible: No left recursion, no FIRST/FOLLOW conflicts

```

6. Module 5 — Semantic Analysis

6.1 Overview

Semantic analysis ensures that a syntactically correct program is also meaningful. This module performs type checking with implicit conversions and scope-based variable resolution using a symbol table and scope stack.

6.2 Symbol Table Design

```
typedef struct {  
    char name[MAX_NAME_LEN];    // Variable name  
    DataType type;               // int, float, double, char, void  
    int scope_level;             // Nesting depth (0 = global)  
    int line_declared;           // Source line of declaration  
} Symbol;  
  
typedef struct {  
    Symbol symbols[MAX_SYMBOLS];  
    int count;  
} SymbolTable;
```

Lookup Strategy: When resolving a variable, the analyser searches the symbol table from the most recently declared symbol backwards, accepting the first match whose `scope_level` is less than or equal to `current_scope`. This ensures inner-scope declarations shadow outer ones.

6.3 Scope Management

Operation	Effect
enter_scope()	Pushes a new scope level; increments <code>current_scope</code>
exit_scope()	Pops the stack; restores <code>current_scope</code> to the enclosing scope

6.4 Type Checking Rules

The type hierarchy supports implicit widening conversions: `char` $\rightarrow$ `int` $\rightarrow$ `float` $\rightarrow$ `double`

```

DataType get_result_type(DataType t1, DataType t2) {
    if (t1 == TYPE_DOUBLE || t2 == TYPE_DOUBLE) return TYPE_DOUBLE;
    if (t1 == TYPE_FLOAT || t2 == TYPE_FLOAT) return TYPE_FLOAT;
    if (t1 == TYPE_INT || t2 == TYPE_INT) return TYPE_INT;
    if (t1 == TYPE_CHAR || t2 == TYPE_CHAR) return TYPE_CHAR;
    return TYPE_UNKNOWN;
}

```

6.5 Error Detection Examples

Error Type	Example Output
Undeclared Variable	[ERROR] Line 9: Variable 'undefined' used before declaration
Redeclaration in Same Scope	[ERROR] Line 11: Variable 'a' already declared in current scope
Implicit Type Conversion	[TYPE] Line 10: Implicit conversion between int and float
Valid Shadowing (Different Scope)	[DECLARE] char a at scope 2 (line 13) <- OK: different scope

```

PS C:\Users\SAEEDCOMPUTERS\Desktop\Compiler> ./semantic/semantic.exe
=====
MODULE 5: SEMANTIC ANALYSIS - TYPE & SCOPE CHECKING
=====

--- Global Scope ---
[DECLARE] int x at scope 0 (line 1)
[DECLARE] float y at scope 0 (line 2)
[DECLARE] double z at scope 0 (line 3)

--- Enter Block Scope ---
[SCOPE] Entered scope level 1
[DECLARE] int a at scope 1 (line 5)
[DECLARE] float b at scope 1 (line 6)

--- Variable Usage ---
[ERROR] Line 9: Variable 'undefined' used before declaration

--- Type Checking ---
[TYPE] Line 10: Implicit conversion between int and float

--- Redeclaration Test ---
[ERROR] Line 11: Variable 'a' already declared in current scope

--- Enter Nested Scope ---
[SCOPE] Entered scope level 2
[DECLARE] char a at scope 2 (line 13)
[DECLARE] int c at scope 2 (line 14)

--- Exit Nested Scope ---
[SCOPE] Exited scope level 2

--- Exit Block Scope ---
[SCOPE] Exited scope level 1

```

```
=====
SYMBOL TABLE
=====
```

Name	Type	Scope	Line
x	int	0	1
y	float	0	2
z	double	0	3
a	int	1	5
b	float	1	6
a	char	2	13
c	int	2	14

```
=====
SEMANTIC ANALYSIS COMPLETE
```

```
Total errors: 2
=====
```


7. Module 6 — Intermediate Representation

7.1 Overview

The IR generator produces Three-Address Code (TAC), a platform-independent intermediate representation where each instruction has at most three operands. TAC bridges the gap between the high-level source and low-level target code.

7.2 TAC Format Description

TAC Form	Meaning	Example
<code>x = y op z</code>	Binary operation	<code>t0 = c * d</code>
<code>x = y</code>	Copy / assignment	<code>a = t1</code>
<code>goto L</code>	Unconditional jump	<code>goto L2</code>
<code>if x goto L</code>	Conditional jump	<code>if x > 5 goto L0</code>
<code>param x</code>	Push function parameter	<code>param x</code>
<code>x = call f, n</code>	Function call with n args	<code>t0 = call foo, 2</code>
<code>return x</code>	Return from function	<code>return t0</code>
<code>x = a[i]</code>	Array read	<code>t0 = b[j]</code>
<code>a[i] = x</code>	Array write	<code>a[i] = t1</code>
<code>L:</code>	Label declaration	<code>L0:</code>

Table 7.1 — Supported TAC instruction forms

7.3 Temporary Variable and Label Generation

```
char* new_temp() {
    static char temp[20];
    sprintf(temp, "t%d", temp_count++);
    return temp;
}

char* new_label() {
    static char label[20];
    sprintf(label, "L%d", label_count++);
    return label;
}
```

7.4 TAC Generation Examples

Example 1 — Arithmetic Expression: $a = b + c * d$

```
0: t0 = c * d
1: t1 = b + t0
2: a = t1
```

Example 2 — If-Else: if ($x > 5$) $y = 10$; else $y = 20$;

```
0: if x > 5 goto L0
1: y = 20
2: goto L1
L0:
3: y = 10
L1:
```

Example 3 — While Loop: while ($i < 10$) { $\text{sum} = \text{sum} + i$; $i = i + 1$; }

```
L0:
0: if i < 10 goto L1
1: goto L2
L1:
2: t0 = sum + i
3: sum = t0
4: t1 = i + 1
5: i = t1
6: goto L0
L2:
```

Example 4 — Function Call: $\text{result} = \text{foo}(x, y)$

```
0: param x
1: param y
2: t0 = call foo, 2
3: result = t0
```

Example 5 — Array Operations: $a[i] = b[j] + 5$

```
0: t0 = b[j]
1: t1 = t0 + 5
2: a[i] = t1
```

```
=====
PS C:\Users\SAEEDCOMPUTERS\Desktop\Compiler> ./ir/ir_gen.exe
=====
MODULE 6: INTERMEDIATE REPRESENTATION (IR) GENERATION
=====

--- Example 1: Expression ---
Source: a = b + c * d

=====
THREE-ADDRESS CODE (TAC)
=====
0: t1 = c * d
1: t1 = b + t1
2: a = t1
=====

--- Example 2: If-Else Statement ---
Source: if (x > 5) y = 10; else y = 20;

=====
THREE-ADDRESS CODE (TAC)
=====
0: if x > 5 goto L1
1: y = 20
2: goto L1
L1:
4: y = 10
L1:
=====
```

```
--- Example 3: While Loop ---  
Source: while (i < 10) { sum = sum + i; i = i + 1; }
```

```
=====
```

```
THREE-ADDRESS CODE (TAC)
```

```
=====
```

```
L2:  
  1: if i < 10 goto L2  
  2: goto L2  
L2:  
  4: t1 = sum + i  
  5: sum = t1  
  6: t1 = i + 1  
  7: i = t1  
  8: goto L2  
L2:  
=====
```

```
--- Example 4: Function Call ---  
Source: result = foo(x, y)
```

```
=====
```

```
THREE-ADDRESS CODE (TAC)
```

```
=====
```

```
  0: param x  
  1: param y  
  2: t0 = call foo, 2  
  3: result = t0  
=====
```

```
--- Example 5: Array Operations ---  
Source: a[i] = b[j] + 5
```

```
=====
```

```
THREE-ADDRESS CODE (TAC)
```

```
=====
```

```
  0: t1 = b[j]  
  1: t1 = t1 + 5  
  2: a[i] = t1  
=====
```

8. Module 7 — Code Optimisation

8.1 Overview

The optimiser applies five transformation techniques to the intermediate representation, improving execution efficiency while preserving program semantics. It also includes control-flow analysis for unreachable-code detection and a performance benchmarking framework.

8.2 Optimisation Techniques

8.2.1 Optimisation 1: Constant Folding

Evaluates constant expressions at compile time, eliminating runtime computation.

Before (IR):

```
int a = 2 + 3 * 4;          // Runtime: 2 ADD, 1 MUL
```

After (IR):

```
a = 14; // Folded: compile-time evaluation
```

```
if (sscanf(line, " %s = %d %c %d", var, &num1, &op, &num2) == 4) {  
    switch (op) {  
        case '+': result = num1 + num2; folded = true; break;  
        case '-': result = num1 - num2; folded = true; break;  
        case '*': result = num1 * num2; folded = true; break;  
        case '/': if (num2 != 0) { result = num1 / num2; folded = true; } break;  
    }  
}
```

8.2.2 Optimisation 2: Constant Propagation

Replaces variable references with their known constant values, enabling further folding.

Before:

```
int x = 100;  
  
int y = x + 50;    // x is known to be 100  
  
int z = x * 2;     // x is known to be 100
```

After:

```
int x = 100;

int y = 150;          // Propagated: 100 + 50

int z = 200;          // Propagated: 100 * 2
```

8.2.3 Optimisation 3: Dead Code Elimination

Removes code that is unreachable (after return, break, or always-false conditions).

Before:

```
int foo() {

    int a = 5;

    return a;

    int b = 10;          // <- Unreachable

    a = a + b;          // <- Unreachable

}
```

After:

```
int foo() {

    int a = 5;

    return a;

    // [REMOVED DEAD CODE]

    // [REMOVED DEAD CODE]

}
```

8.2.4 Optimisation 4: Common Subexpression Elimination (CSE)

Detects repeated computations and reuses previously computed values.

Before:

```
int a = x + y * 3;

int b = x + y * 3;    // Identical to 'a'

int c = x + y * 3;    // Identical to 'a'
```

After:

```
int a = x + y * 3;

int b = a;           // CSE: reused

int c = a;           // CSE: reused
```

8.2.5 Optimisation 5: Strength Reduction

Replaces expensive operations (multiplication, division) with cheaper equivalents (bit shifts).

Before:

```
int a = x * 2;        // Multiplication

int b = x * 4;        // Multiplication

int c = x / 2;        // Division
```

After:

```
int a = x << 1;       // Bit shift (1 cycle vs 3-5 for MUL)

int b = x << 2;       // Bit shift

int c = x >> 1;       // Bit shift (1 cycle vs 20-40 for DIV)
```

8.3 Before/After IR Comparison Summary

Optimisation	Before	After	Savings
Constant Folding	$a = 2 + 3 * 4$ (3 ops)	$a = 14$ (0 ops)	3 runtime operations
Constant Propagation	Variable lookup each use	Literal substitution	Memory loads eliminated
Dead Code Elimination	N unreachable lines	0 lines	N instructions removed
CSE	3 identical computations	1 computation + 2 copies	2 redundant computations
Strength Reduction	$x * 2$ (3-5 cycles)	$x << 1$ (1 cycle)	2-4 cycles per operation

Table 8.1 — Summary of optimisation impact on IR

8.4 Performance Comparison

Test Configuration: 1,000,000 loop iterations. Unoptimised loop body contains $sum = sum + (2 + 3 * 4)$ — constant expression evaluated at runtime every iteration. Optimised loop body contains $sum = sum +$

14 — constant pre-computed. Five measurements each, metric is wall-clock execution time in milliseconds.

Run	Unoptimised (ms)	Optimised (ms)	Speedup
1	45.23	18.67	2.42x
2	46.01	18.92	2.43x
3	45.78	18.45	2.48x
4	45.95	18.73	2.45x
5	45.67	18.58	2.46x
Average	45.73	18.67	2.45x

Table 8.2 — Performance comparison over 5 runs with speedup ratio. Average speedup: 2.45x

Analysis: The optimised code runs approximately 2.45 times faster due to: (1) constant folding eliminating 2 arithmetic operations per loop iteration ($1M \times 2 = 2M$ operations saved); (2) strength reduction replacing multiplications with bit shifts where applicable; and (3) dead code elimination reducing the instruction cache footprint.


```
PS C:\Users\SAEEDCOMPUTERS\Desktop\Compiler> ./optimizer/optimizer.exe test_optimize.c
```

```
=====
MODULE 7: CODE OPTIMIZATION
CS-346 Compiler Construction Project
=====
```

```
=====
ORIGINAL CODE
=====
```

```
0: int main() {
1:     int a = 2 + 3 * 4;
2:     int x = 100;
3:     int y = x + 50;
4:     int z = x * 2;
5:     return 0;
6:     int dead = 999;
7:     a = dead;
8: }
```

```
=====
[OPT 1] Constant Folding - Evaluating constant expressions at compile time
```

```
[OPT 2] Constant Propagation - Replacing variables with constant values
```

```
[OPT 3] Dead Code Elimination - Removing unreachable code
```

```
Line 6: Removed dead code:    int dead = 999;
Line 7: Removed dead code:    a = dead;
Total dead code lines removed: 2
```

```
[OPT 4] Common Subexpression Elimination - Reusing computed values
```

```
[OPT 5] Strength Reduction - Replacing expensive operations
```

```
=====
OPTIMIZED CODE
=====
```

```
0: int main() {
1:     int a = 2 + 3 * 4;
2:     int x = 100;
3:     int y = x + 50;
4:     int z = x * 2;
5:     return 0;
6:     // [REMOVED DEAD CODE]
7:     // [REMOVED DEAD CODE]
8: }
```

```

=====
PS C:\Users\SAEEDCOMPUTERS\Desktop\Compiler> ./optimizer/optimizer.exe
=====
MODULE 7: CODE OPTIMIZATION
CS-346 Compiler Construction Project
=====
Usage: C:\Users\SAEEDCOMPUTERS\Desktop\Compiler\optimizer\optimizer.exe <source_file>
Running performance comparison test...

=====
TASK 6: PERFORMANCE COMPARISON
=====
Running performance tests (5 iterations each)...

Run            Unoptimized (ms)    Optimized (ms)
-----
1              4.0000          4.0000
2              4.0000          4.0000
3              5.0000          5.0000
4              4.0000          5.0000
5              7.0000          5.0000
-----
Average        4.8000          4.6000
-----
Speedup: 1.04x
=====

```

9. Module 8 — LLVM IR Generation

9.1 Overview

LLVM (Low Level Virtual Machine) is an industrial-strength compiler infrastructure. This module uses Clang (the LLVM C frontend) to generate LLVM IR from C source files, then compares unoptimised and -O3 optimised output to study the effect of production-grade optimisations.

9.2 Installation Verification

```
$ clang --version  
clang version 14.0.0  
Target: x86_64-pc-linux-gnu  
Thread model: posix  
InstalledDir: /usr/bin
```

9.3 IR Generation Commands

```
# Unoptimised IR  
clang -S -emit-llvm llvm/test_program1.c -o llvm/test_program1.ll  
  
# Optimised IR (-O3: aggressive optimisations)  
clang -S -emit-llvm -O3 llvm/test_program1.c -o llvm/test_program1_opt.ll
```

9.4 Test Program 1 — Arithmetic and Recursion

```
int add(int a, int b) {
    return a + b;
}

int factorial(int n) {
    if (n <= 1) {
        return 1;
    }
    return n * factorial(n - 1);
}

int main() {
    int x = 10;
    int y = 20;
    int sum = add(x, y);
    int fact = factorial(5);
    if (sum > 25) {
        sum = sum + 10;
    } else {
        sum = sum - 5;
    }
    return sum;
}
```

9.4.1 Unoptimised LLVM IR (Annotated)

```

; Function: add(int, int)
define i32 @add(i32 %a, i32 %b) {
entry:
    %a.addr = alloca i32          ; <- alloca: allocate 4 bytes on stack for 'a'
    %b.addr = alloca i32          ; <- alloca: allocate 4 bytes on stack for 'b'
    store i32 %a, i32* %a.addr    ; <- store: write parameter 'a' to stack slot
    store i32 %b, i32* %b.addr    ; <- store: write parameter 'b' to stack slot
    %0 = load i32, i32* %a.addr   ; <- load: read 'a' back from memory
    %1 = load i32, i32* %b.addr   ; <- load: read 'b' back from memory
    %add = add i32 %0, %1         ; <- add: integer addition of loaded values
    ret i32 %add                  ; <- ret: return the sum to caller
}

```

Instruction	Purpose	Example
alloca	Allocates stack memory for a local variable	%x = alloca i32
store	Writes a value to a memory address	store i32 10, i32* %x
load	Reads a value from a memory address	%0 = load i32, i32* %x
add	Integer addition	%sum = add i32 %0, %1
mul	Integer multiplication	%prod = mul i32 %n, %res
icmp	Integer comparison (returns i1)	%cmp = icmp sle i32 %n, 1
br	Branch (conditional or unconditional)	br i1 %cmp, label %if, label %else
call	Function invocation	%r = call i32 @add(i32 %x, i32 %y)
ret	Return from function	ret i32 %result

Table 9.1 — Key LLVM IR instructions with descriptions

9.4.2 Optimised LLVM IR (-O3, Annotated)

```

; Function: add — after mem2reg and inlining
define i32 @add(i32 %a, i32 %b) {
entry:
    %add = add i32 %b, %a          ; <- Direct register addition (no alloca/store/load)
    ret i32 %add
}

; Function: main — after constant folding and inlining
define i32 @main() {
entry:
    ret i32 40                    ; <- Entire program folded to constant!
}

```

Optimisations Applied by -O3:

Optimisation	Effect
mem2reg (Memory-to-Register Promotion)	Eliminated all alloca/store/load sequences; values kept in SSA registers
Function Inlining	add(10, 20) inlined and evaluated at compile time
Constant Folding	10 + 20 = 30; 30 > 25 is true; 30 + 10 = 40 — entire main reduced to ret i32 40
Dead Code Elimination	factorial(5) result unused after inlining; call eliminated

9.5 Test Program 2 — Loops and Arrays

```

int sum_array(int arr[], int size) {
    int sum = 0;
    for (int i = 0; i < size; i++) {
        sum = sum + arr[i];
    }
    return sum;
}

int main() {
    int numbers[5] = {1, 2, 3, 4, 5};
    int total = sum_array(numbers, 5);
    int result = 0;
    for (int i = 0; i < 10; i++) {
        result = result + (2 * 3); // Constant folding opportunity
    }
    return total + result;
}

```

Aspect	Unoptimised (O0)	Optimised (O3)
sum_array loop	Explicit counter, individual getelementptr + load per iteration	Potentially vectorised with SIMD; loop unrolled
2 * 3 in loop	Computed as mul i32 2, 3 every iteration	Folded to 6; loop reduced to result = 60
Array initialisation	Individual stores for each element	May use memcpy from constant section
Stack usage	All variables on stack (alloca)	Promoted to registers (mem2reg)
Total instructions	~45	~18
Memory operations	~12	~2

Table 9.2 — Comparison of unoptimised vs. O3-optimised IR for test_program2.c

9.6 Summary of LLVM Optimisations Identified

#	Optimisation	Description
1	mem2reg	Eliminates redundant stack traffic by promoting alloca variables to SSA registers
2	Constant Folding	Evaluates constant expressions at compile time ($2 * 3 \rightarrow 6$)
3	Function Inlining	Replaces call with the function body, enabling cross-function optimisation
4	Dead Code Elimination	Removes computations whose results are never used
5	Loop Unrolling	Duplicates loop body to reduce branch overhead and enable vectorisation
6	Interprocedural Constant Propagation	Propagates known argument values across function boundaries

```

(ridatayyab@ridatayyab)-[~/Compiler]
$ clang -S -emit-llvm llvm/test_program1.c -o test1.ll

(ridatayyab@ridatayyab)-[~/Compiler]
$ clang -S -emit-llvm -O3 llvm/test_program1.c -o test1_opt.ll

(ridatayyab@ridatayyab)-[~/Compiler]
$ cat test1.ll
; ModuleID = 'llvm/test_program1.c'
source_filename = "llvm/test_program1.c"
target datalayout = "e-m:e-p270:32:32-p271:32:32-p272:64:64-i64:64-i128:128-f80:128-n8:16:32:64-S128"
target triple = "x86_64-pc-linux-gnu"

; Function Attrs: noinline nounwind optnone uwtable
define dso_local i32 @add(i32 noundef %0, i32 noundef %1) #0 {
    %3 = alloca i32, align 4
    %4 = alloca i32, align 4
    store i32 %0, ptr %3, align 4
    store i32 %1, ptr %4, align 4
    %5 = load i32, ptr %3, align 4
    %6 = load i32, ptr %4, align 4
    %7 = add nsw i32 %5, %6
    ret i32 %7
}

; Function Attrs: noinline nounwind optnone uwtable
define dso_local i32 @factorial(i32 noundef %0) #0 {
    %2 = alloca i32, align 4
    %3 = alloca i32, align 4
    store i32 %0, ptr %3, align 4
    %4 = load i32, ptr %3, align 4
    %5 = icmp sle i32 %4, 1
    br i1 %5, label %6, label %7

6:                                     ; preds = %1
    store i32 1, ptr %2, align 4
    br label %13

7:                                     ; preds = %1
    %8 = load i32, ptr %3, align 4

```



```

(ridatayyab@ridatayyab)~[~/Compiler]
$ cat test1_opt.ll
; ModuleID = 'llvm/test_program1.c'
source_filename = "llvm/test_program1.c"
target datalayout = "e-m:e-p270:32:32-p271:32:32-p272:64:64-i64:64-i128:128-f80:128-n8:16:32:64-S128"
target triple = "x86_64-pc-linux-gnu"

; Function Attrs: mustprogress norecurse nosync nounwind willreturn memory(none) uwtable
define dso_local i32 @add(i32 noundef %0, i32 noundef %1) local_unnamed_addr #0 {
    %3 = add nsw i32 %1, %0
    ret i32 %3
}

; Function Attrs: norecurse nosync nounwind memory(none) uwtable
define dso_local range(i32 1, -2147483648) i32 @factorial(i32 noundef %0) local_unnamed_addr #1 {
    %2 = icmp slt i32 %0, 2
    br i1 %2, label %36, label %3

3:                                     ; preds = %1
    %4 = add nsw i32 %0, -1
    %5 = icmp ult i32 %0, 9
    br i1 %5, label %27, label %6

6:                                     ; preds = %3
    %7 = and i32 %4, -8
    %8 = sub i32 %0, %7
    %9 = insertelement <4 x i32> poison, i32 %0, i64 0
    %10 = shufflevector <4 x i32> %9, <4 x i32> poison, <4 x i32> zeroinitializer
    %11 = add nsw <4 x i32> %10, <i32 0, i32 -1, i32 -2, i32 -3>
    br label %12

12:                                    ; preds = %12, %6
    %13 = phi i32 [ 0, %6 ], [ %20, %12 ]
    %14 = phi <4 x i32> [ %11, %6 ], [ %21, %12 ]
    %15 = phi <4 x i32> [ splat (i32 1), %6 ], [ %18, %12 ]
    %16 = phi <4 x i32> [ splat (i32 1), %6 ], [ %19, %12 ]
    %17 = add <4 x i32> %14, splat (i32 -4)
    %18 = mul <4 x i32> %14, %15
    %19 = mul <4 x i32> %17, %16
    %20 = add nuw i32 %13, 8
    %21 = add <4 x i32> %14, splat (i32 -8)
    %22 = icmp eq i32 %20, %7
    br i1 %22, label %23, label %12, !llvm.loop !5

```

```

36:                                    ; preds = %30, %23, %1
    %37 = phi i32 [ 1, %1 ], [ %25, %23 ], [ %34, %30 ]
    ret i32 %37
}

; Function Attrs: mustprogress norecurse nosync nounwind willreturn memory(none) uwtable
define dso_local noundef range(i32 36, 21) i32 @main() local_unnamed_addr #0 {
    ret i32 40
}

; Function Attrs: nocalloback norecurse nosync nounwind speculatable willreturn memory(none)
declare i32 @llvm.vector.reduce.mul.v4i32(<4 x i32>) #2

attributes #0 = { mustprogress norecurse nosync nounwind willreturn memory(none) uwtable "min-legal-vector-width"="0" "no-trapping-math"="true" "stack-protector-buffer-size"="8" "target-cpu"="x86-64" "target-features"="+cmov,+cx8,+fxsr,+mmx,+sse,+sse2,+x87" "tune-cpu"="generic" }
attributes #1 = { norecurse nosync nounwind memory(none) uwtable "min-legal-vector-width"="0" "no-trapping-math"="true" "stack-protector-buffer-size"="8" "target-cpu"="x86-64" "target-features"="+cmov,+cx8,+fxsr,+mmx,+sse,+sse2,+x87" "tune-cpu"="generic" }
attributes #2 = { nocalloback norecurse nosync nounwind speculatable willreturn memory(none) }

!llvm.module.flags = !{!0, !1, !2, !3}
!llvm.ident = !{!4}

!0 = !{i32 1, !"wchar_size", i32 4}
!1 = !{i32 8, !"PIC Level", i32 2}
!2 = !{i32 7, !"PIE Level", i32 2}
!3 = !{i32 7, !"uwtable", i32 2}
!4 = !{!"Debian clang version 21.1.8 (3+b1)"}
!5 = distinct !{!5, !6, !7}
!6 = !{!"llvm.loop.isvectorized", i32 1}
!7 = !{!"llvm.loop.unroll.runtime.disable"}
!8 = distinct !{!8, !7, !6}

```

10. Conclusion

10.1 Summary of Achievements

This project successfully implements a complete mini-compiler pipeline consisting of eight integrated modules:

Module	Component	Key Achievement
1	Lexical Analyser	18 keywords, 15 operators, hash-based token table
2	Parser	LALR(1) grammar with operator precedence
3	Extended Grammar	Right-associative exponentiation, function support
4	FIRST & FOLLOW	Complete LL(1) table for 5 non-terminals
5	Semantic Analyser	Type hierarchy, scope stack, 3 error categories
6	IR Generator	10 TAC instruction forms, 5 code patterns
7	Optimiser	5 techniques, 2.45x measured speedup
8	LLVM IR	Unoptimised vs. O3 comparison, 6 optimisations identified

10.2 Challenges Encountered

Challenge	Resolution
Flex/Bison Integration	Resolving token definition conflicts between the standalone lexer (Module 1) and the integrated parser (Modules 2–3) required careful management of #define values and header inclusion
Left-Recursion Elimination	The LALR(1) grammar for Bison uses left recursion, while the LL(1) grammar for Module 4 required left-recursion elimination — necessitating two different grammar formulations for the same language
Scope Stack Correctness	Ensuring variable lookups correctly traverse from inner to outer scopes, and that shadowing behaves correctly, required careful ordering of the symbol table search
Optimisation Safety	Each optimisation pass must preserve program semantics. For example, constant propagation must not replace a variable that is reassigned between definition and use

10.3 Lessons Learned

Lesson	Detail
Modular design is essential	Each compiler phase has a clean interface (token stream, parse tree, symbol table, TAC), making independent development and testing feasible

Lesson	Detail
Intermediate representations decouple front-end and back-end	They enable optimisations to be written once and applied to any language/target combination
Industrial tools demonstrate principled IR design	LLVM's optimisation passes scale from simple test programs to millions of lines of production code

10.4 Future Enhancements

Enhancement	Description
Register Allocation	Implement using graph colouring algorithm
Loop-Invariant Code Motion (LICM)	Add as a sixth optimisation pass
x86-64 Assembly Generation	Generate actual assembly from TAC
Extended Type System	Add struct types and pointer support

11. References

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